CHAPTER 08 / SUPPLEMENTS

Supplements, explained

Evidence before promises.

Understand what helps, what is optional and what is uncertain.

How to use this guide

Read the explanations, try the practical examples and complete the worksheet. All training logs and menus are illustrative. Use the website for interactive tools; this document remains useful offline.

01	02	03
Understand	Put into practice	Review

Scope and safety

General education for adults, not a medical diagnosis, personalized treatment or a guaranteed transformation. Stop exercise for sharp pain or unusual symptoms. A qualified clinician, dietitian or coach can help when individual circumstances require assessment.

What you will learn

01. Start with the foundation

02. Separate the claims

03. Ask the right questions

01 / Start with the foundation

Supplements add to a diet; they cannot substitute for consistent training, sleep and enough food. Product quality varies. Prefer transparent labels and independent batch testing where available, especially in tested sport. Marketing phrases are not evidence.

PUT IT INTO PRACTICE: Before buying, name the specific problem you want the product to solve.

02 / Separate the claims

A substance can correct a deficiency without improving muscle growth in someone who already has enough. Evidence for an ingredient is not proof that every product or blend works. Check the population, outcome and limitations of a study.

PUT IT INTO PRACTICE: An endurance result does not automatically demonstrate a hypertrophy benefit.

03 / Ask the right questions

Consider allergies, medication interactions, existing conditions and total intake from other products. A clinician or pharmacist can help when you have a condition, take medication or are pregnant. Do not treat this educational guide as a personalized supplement prescription.

PUT IT INTO PRACTICE: Bring the full ingredient list when asking for professional advice.

Practical reference

Whey protein

Evidence context	Strong for meeting protein needs
Potential role	A convenient milk-derived protein; it can help when food intake falls short.
Limitations	Not superior to an adequate overall diet.
Safety / individual advice	Milk allergy and digestive intolerance are relevant. Optional; check suitability with a clinician for kidney disease or a medically restricted diet.

Creatine monohydrate

Evidence context	Strong for repeated high-intensity performance
Potential role	Can support repeated high-intensity efforts and training-related lean-mass gains.
Limitations	Not a substitute for training; early scale gain can include water.
Safety / individual advice	Possible digestive discomfort and water-related weight gain. Optional; discuss kidney conditions, pregnancy or medication concerns with a clinician.

Caffeine

Evidence context	Evidence for performance; response varies
Potential role	Can improve alertness and some exercise performance.
Limitations	More is not better, and sleep disruption can undermine recovery.
Safety / individual advice	May cause anxiety, palpitations or insomnia. Avoid concentrated powders and stimulant stacking; seek advice with heart conditions, pregnancy or interacting medicines.

Omega-3

Evidence context	Limited or mixed for muscle growth
Potential role	Provides fatty acids found in oily fish and some other foods.
Limitations	A direct muscle-building effect is not established for every healthy lifter.
Safety / individual advice	Digestive effects and medication interactions matter, especially with anticoagulants. Discuss supplementation if your diet is limited.

Vitamin D

Evidence context	Useful for deficiency; not a universal muscle booster
Potential role	Supports normal bone and mineral regulation.
Limitations	Extra intake does not guarantee extra muscle when status is adequate.
Safety / individual advice	Excess can be harmful. Assessment and an appropriate plan are best guided by a clinician when deficiency is suspected.

Multivitamins

Evidence context	Useful for selected nutrient gaps
Potential role	May help cover specific dietary shortfalls.
Limitations	Cannot replace varied food or guarantee improved training results.
Safety / individual advice	Avoid overlapping high-dose products; check interactions and individual needs with a professional.

Electrolytes

Evidence context	Context-dependent
Potential role	Can help replace salts lost during prolonged heavy sweating.
Limitations	Often unnecessary for a short session with normal meals and water.
Safety / individual advice	Excess or inappropriate use may be problematic with kidney, heart or blood-pressure conditions; individualized advice may be needed.

Your study worksheet

Knowledge check

A supplement with benefits for deficiency…

A. Must build more muscle in everyone

B. May not help someone already sufficient

C. Replaces a balanced diet

Reflect and apply

Prompt	Your response
One idea I can explain in my own words	
One action I can try this week	
What I will observe or measure	
What would make me seek professional guidance	
My next review date	

Chapter summary

Treat supplements as optional, match evidence to the claim and consider individual risks.

Answer and explanation

B. Correcting a shortfall and enhancing performance are different claims.

Sources & limitations

These sources inform the educational principles. Original examples and planning templates are not study findings. No institution listed below endorses this journal. References checked November 2026.

NIH Office of Dietary Supplements — Exercise and athletic performance

https://ods.od.nih.gov/factsheets/ExerciseAndAthleticPerformance-HealthProfessional/

NIH Office of Dietary Supplements — Vitamin D

https://ods.od.nih.gov/factsheets/VitaminD-HealthProfessional/

NIH Office of Dietary Supplements — Omega-3 fatty acids

https://ods.od.nih.gov/factsheets/Omega3FattyAcids-HealthProfessional/

NIH Office of Dietary Supplements — Multivitamin and mineral supplements

https://ods.od.nih.gov/factsheets/MVMS-Consumer/

ACSM — Exercise and fluid replacement

https://pubmed.ncbi.nlm.nih.gov/17277604/

Interpretation matters

Research describes groups under particular conditions. Individual responses vary. The website models are simplified illustrations, not precise physiological simulations. Food values and calorie estimates are approximate. Use consistent observation and individualized professional support when needed.